# Problem

Stewart, *Calculus: Early Transcendentals*, Section 15.7 (Triple Integrals), Exercise 27.

Sketch the solid whose volume is given by the iterated integral

$$\int_0^1 \int_0^{1-x} \int_0^{2-2z} dy\, dz\, dx.$$

The solution must identify the bounding planes, describe the solid precisely, draw it, and compute the volume as a check.

# Part (a)

## Notation

The symbol $E$ denotes the solid region in space, that is, the set of points $(x, y, z)$ whose volume the integral measures. The symbol $dV$ denotes the volume element. When the differentials are written in the order $dy\, dz\, dx$, the volume element is

$$dV = dy\, dz\, dx.$$

A triple integral $\iiint_E f(x, y, z)\, dV$ adds the values of $f$ over $E$ together with the volume weights $dV$. An *iterated integral* evaluates the same object as three nested single integrals. Fubini's theorem states that for a bounded solid $E$ and a continuous function $f$ the iterated integral exists and equals the triple integral, for every order of integration that fits the geometry of $E$.

## Step 1: read the three pairs of limits

An iterated integral is read from the innermost pair of limits outward. The innermost pair belongs to the differential that stands closest to the integrand.

- Innermost, $dy$: the limits are 0 and $2 - 2z$. They say that for fixed $x$ and $z$ the variable $y$ obeys
$$0 \le y \le 2 - 2z.$$
- Middle, $dz$: the limits are 0 and $1 - x$. They say that for fixed $x$ the variable $z$ obeys
$$0 \le z \le 1 - x.$$

- Outermost, $dx$: the limits are 0 and 1. They say that

$$0 \le x \le 1.$$

The double inequality $0 \le y \le 2 - 2z$ has two parts. The lower part $y = 0$ is a plane. The upper part $y = 2 - 2z$ is also a plane, namely $y + 2z = 2$. In the same way the pair $0 \le z \le 1 - x$ gives the plane $z = 0$ and the plane $x + z = 1$.

### Step 2: the solid described by the three inequalities

The three pairs of limits are equivalent to the single description

$$E = \{(x, y, z) : 0 \le x \le 1,\ 0 \le z \le 1 - x,\ 0 \le y \le 2 - 2z\}.$$

Read the description in two stages. First look at $x$ and $z$ alone. The conditions $0 \le x \le 1$ and $0 \le z \le 1 - x$ say that the pair $(x, z)$ lies in the triangle

$$T = \{(x, z) : x \ge 0,\ z \ge 0,\ x + z \le 1\},$$

with vertices $(0, 0)$, $(1, 0)$ and $(0, 1)$. Second, over every point of $T$ the variable $y$ runs from the plane $y = 0$ up to the plane $y + 2z = 2$.

The triangle $T$ is the shadow of $E$ on the plane $y = 0$. The solid $E$ is the part of the space above $T$ that is cut off by the oblique plane $y + 2z = 2$.

The upper limit of $x$ is 1. The plane $x = 1$ is not a face of $E$. At $x = 1$ the condition $0 \le z \le 1 - x$ forces $z = 0$, so the plane $x = 1$ touches $E$ only along the single edge from $(1, 0, 0)$ to $(1, 2, 0)$. The number 1 is therefore only the largest value that $x$ takes on $T$.

### Step 3: the five faces and the vertices

Every pair of limits that can hold with equality supplies a face of $E$. Table 1 lists the five faces. Each side face is a triangle, and each side face contains the point $(0, 0, 1)$. The base is a rectangle.

Table 1: The five faces of the solid $E$ of Exercise 27.

| Face | Plane | Vertices | Shape |
|---|---|---|---|
| base | $z = 0$ | $(0,0,0)$, $(1,0,0)$, $(1,2,0)$, $(0,2,0)$ | rectangle $1 \times 2$ |
| left | $x = 0$ | $(0,0,0)$, $(0,2,0)$, $(0,0,1)$ | triangle |
| front | $y = 0$ | $(0,0,0)$, $(1,0,0)$, $(0,0,1)$ | triangle |
| slant left | $x + z = 1$ | $(1,0,0)$, $(1,2,0)$, $(0,0,1)$ | triangle |
| slant top | $y + 2z = 2$ | $(0,2,0)$, $(1,2,0)$, $(0,0,1)$ | triangle |

The five faces use the five planes

$$z = 0, \qquad x = 0, \qquad y = 0, \qquad x + z = 1, \qquad y + 2z = 2.$$

The solid $E$ is the set of points that lie on the inner side of all five planes. The test point $(0.2, 0.2, 0.2)$ gives $x + z = 0.4 < 1$ and $y + 2z = 0.6 < 2$, so the test point is inside $E$.

The solid has five vertices:

$$(0,0,0), \quad (1,0,0), \quad (1,2,0), \quad (0,2,0), \quad (0,0,1).$$

The last vertex is the apex. It lies on all four side planes: $x = 0$, $y = 0$, $x + z = 1$ and $y + 2z = 2$.

### Step 4: the solid is a pyramid

Let $z$ be a fixed number with $0 \le z \le 1$. The points of $E$ at that height satisfy

$$0 \le x \le 1 - z, \qquad 0 \le y \le 2 - 2z.$$

This set is a rectangle with side lengths $1 - z$ and $2 - 2z$. The two side lengths shrink by the same factor $1 - z$ as $z$ goes from 0 to 1. At $z = 0$ the rectangle is the full base $1 \times 2$. At $z = 1$ the rectangle shrinks to the single point $(0, 0, 1)$. A solid whose parallel sections are similar rectangles that shrink to a point is a pyramid with height 1.

The area of the section at height $z$ is

$$A(z) = (1 - z)(2 - 2z) = 2(1 - z)^2.$$

Figure 1 shows the solid. Figure 2 shows the shadow $T$ and four sections.

### Step 5: the volume, computed twice

The volume is the integral of the constant function 1 over $E$. Integrate first in $y$:

$$V = \int_0^1 \int_0^{1-x} \int_0^{2-2z} 1 \, dy \, dz \, dx = \int_0^1 \int_0^{1-x} (2 - 2z) \, dz \, dx.$$

The inner antiderivative is

$$\int (2 - 2z) \, dz = 2z - z^2,$$

so

$$\int_0^{1-x} (2 - 2z) \, dz = \left[2z - z^2\right]_0^{1-x} = 2(1 - x) - (1 - x)^2.$$

Expand the last expression:

$$2(1 - x) - (1 - x)^2 = 2 - 2x - \left(1 - 2x + x^2\right) = 1 - x^2.$$

The outer integral is therefore

$$V = \int_0^1 (1 - x^2) \, dx = \left[x - \frac{x^3}{3}\right]_0^1 = 1 - \frac{1}{3} = \frac{2}{3}.$$

A second computation uses the sections of Step 4:

$$V = \int_0^1 A(z) \, dz = \int_0^1 2(1 - z)^2 \, dz = \left[-\frac{2(1 - z)^3}{3}\right]_0^1 = \frac{2}{3}.$$

A third computation uses the pyramid formula with base area 2 and height 1:

$$V = \frac{1}{3} \cdot 2 \cdot 1 = \frac{2}{3}.$$

All three computations agree. The verification scripts repeat the result with exact symbolic algebra and with three independent numerical methods (composite Simpson rule 0.666666666667, Gaussian quadrature 0.666666666667, Monte Carlo $0.666447 \pm 0.000471$).

$$\boxed{E = \{(x, y, z) : 0 \le x \le 1, \ 0 \le z \le 1 - x, \ 0 \le y \le 2 - 2z\}, \qquad V = \frac{2}{3}}$$

The solid is the pyramid with base $0 \le x \le 1$, $0 \le y \le 2$ in the plane $z = 0$, apex $(0, 0, 1)$, and bounding planes $z = 0$, $x = 0$, $y = 0$, $x + z = 1$, $y + 2z = 2$.

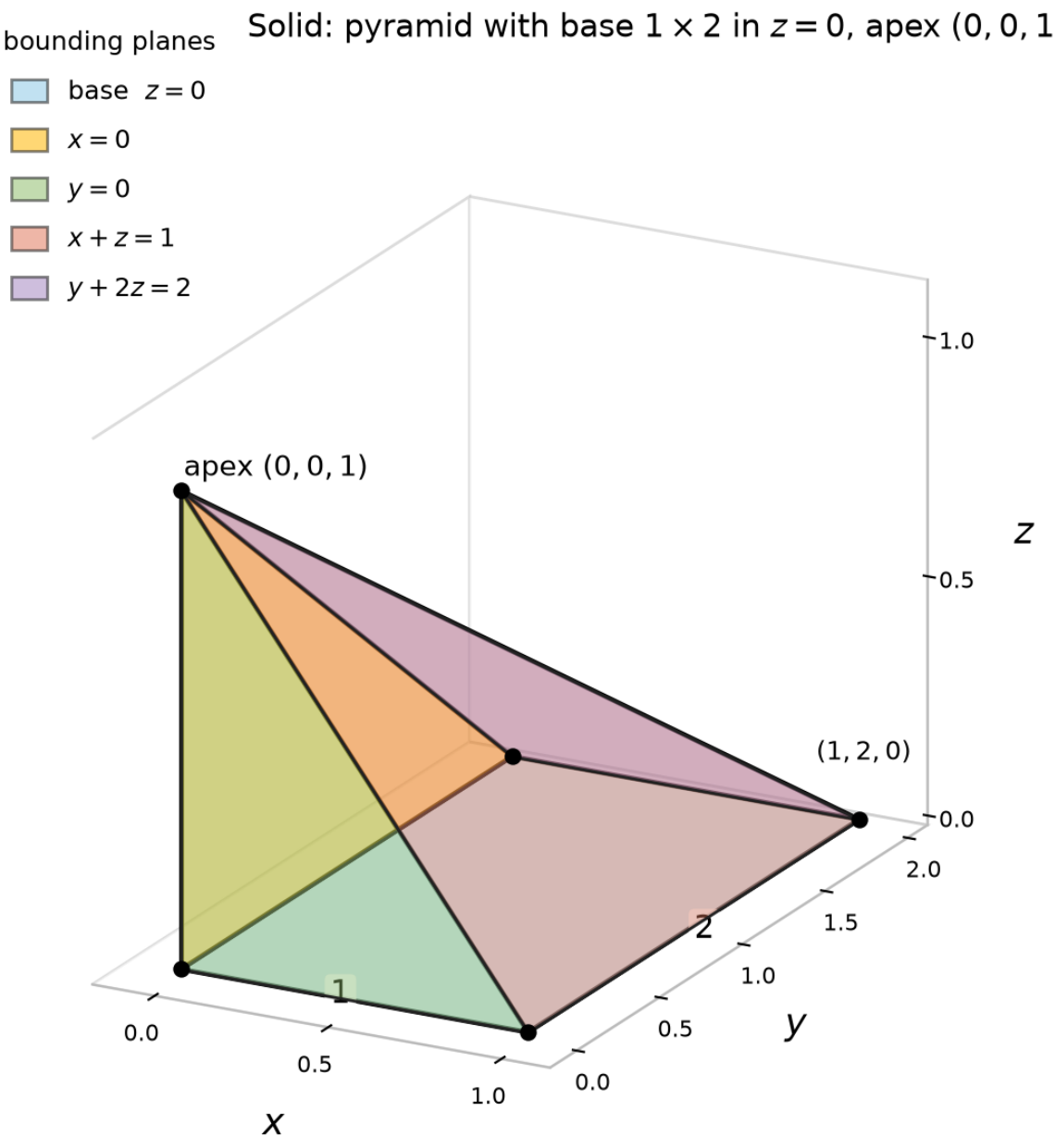



Figure 1: The solid of Exercise 27. The base is the rectangle $0 \le x \le 1$, $0 \le y \le 2$ in the plane $z = 0$. The apex is $(0, 0, 1)$. The five faces are the base and the four planes $x = 0$, $y = 0$, $x + z = 1$ and $y + 2z = 2$. The numbers 1 and 2 are the side lengths of the base; the height is 1.

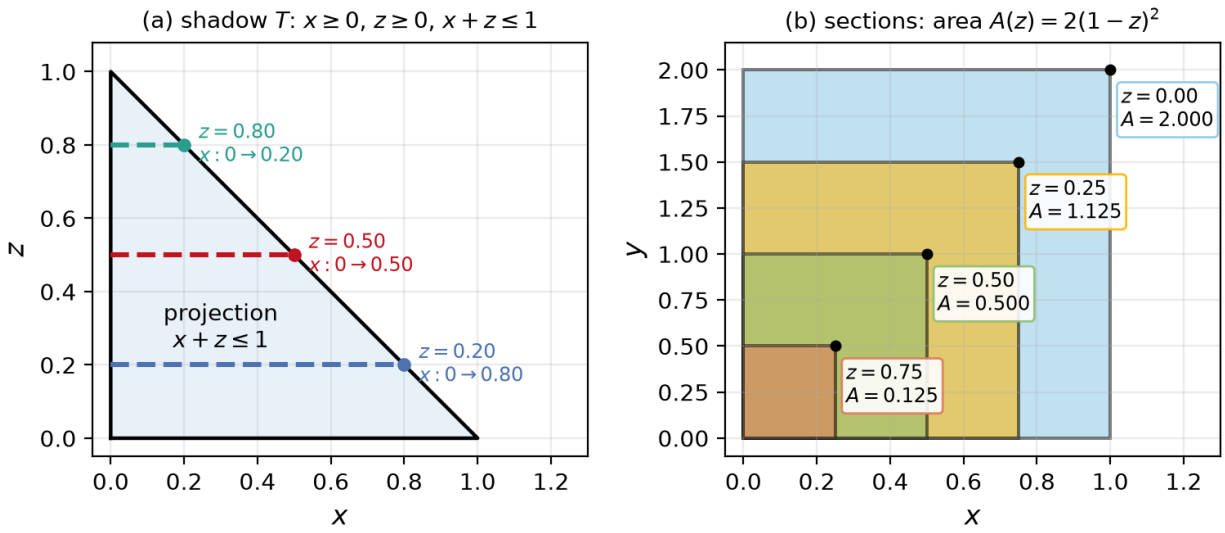



Figure 2: (a) The shadow $T$ of the solid on the plane $y = 0$ is the triangle $x \geq 0$, $z \geq 0$, $x + z \leq 1$. A horizontal line at height $z$ runs from $x = 0$ to $x = 1 - z$. (b) The section of the solid at height $z$ is a rectangle with area $A(z) = 2(1 - z)^2$. The four drawn sections have $z = 0$, 0.25, 0.5 and 0.75.

## How to solve problems of this kind

1. Read the iterated integral from the innermost differential outward. Write one inequality for each pair of limits.
2. Turn every inequality into a surface. An inequality such as $y \leq 2 - 2z$ names the plane $y + 2z = 2$. A pair of equalities such as $y = 0$ names a coordinate plane.
3. Separate the variables into an inner pair and an outer pair. The outer pair describes the shadow of the solid on a coordinate plane. Draw the shadow first; it fixes most of the sketch.
4. Check that each limit depends only on variables that are still free. This condition makes the iterated integral legal.
5. Find the vertices by solving the plane equations in groups. A vertex satisfies at least three plane equations.
6. Take sections that are parallel to a coordinate plane. The area of the section, written as a function of one variable, gives the volume by a single integral.
7. Check the volume with a second method. Use the geometric formula for the shape, a different order of integration, or a numerical method.